



Terminal Approach Production System (TAPS) Version Description Document

Foundation Sustainment Services (FS2)

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Revision History

Version #	Implemented By	Revision Date	Reason
1.0.0	Christi Curtis	03/24/2025	Initial release of this version of the product.

Authorization Notification

Any authorization applies only to the current version of the document.

Role	Name	Company	Job Title
Prepared By			CM Lead / Test Engineer
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Approval

The undersigned have read this VDD and approve its contents.

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Contents

1	Scope	4
1.1	Introduction	4
1.2	System Overview	4
1.3	Document Overview	4
2	Version Description.....	5
2.1	Deployment Environment	5
2.2	Software Functionality and Capabilities	5
2.3	Inventory of Software Components.....	6
2.3.1	Software Deliverables.....	6
2.3.2	Software List.....	6
2.4	Changes Installed	8
2.5	Installation Instructions	8
2.6	Possible Problems and Known Errors	8



1 Scope

1.1 Introduction

This document is the *Version Description Document (VDD) for Terminal Approach Production System TAPS Version 1.0.0*.

This VDD details the software components to be delivered with TAPS Version 1.0.0. This document describes the features included in the delivery of the software build, at entrance to test. It includes the build artifacts for deployment, any known issues with the delivery, and the changes included in the software version.

1.2 System Overview

TAPS is a web-based application that is used for terminal approach production. Users upload data into system and can export filtered data sets for use in OpenCities. The TAPS Server application contains business logic and processing rules. It also contains a robust REST API for use by client programs. The server application runs as a .NET application. The TAPS Web Client application is a single-page web application that provides a user interface for the TAPS system. It interacts with the TAPS Server program to provide the user with functionality and is run from a modern web browser on a client workstation. The TAPS Web Client application is written using Node.js/REACT framework and is deployed using a .NET web server.

1.3 Document Overview

The TAPS VDD provides the list of software delivered for Version 1.0.0 of the product and the capabilities of that software. This document details a single release of the product and will be updated only to correct information specific to version 1.0.0 of the product. Follow-on versions of the product will be documented in a separate VDD specific to that version of the product.

The document is organized as follows:

- Section 1 – Scope
- Section 2 – Version Description



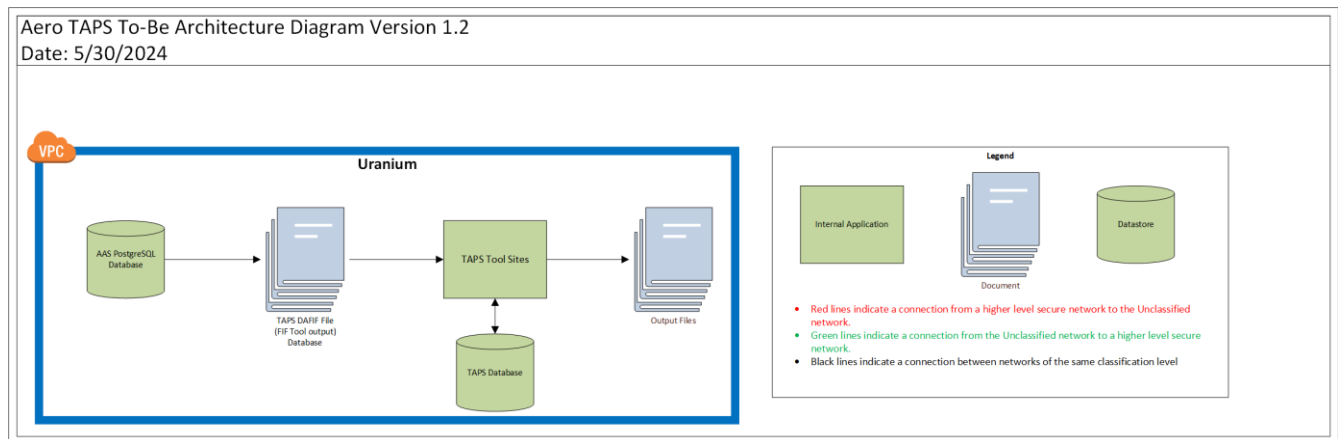
2 Version Description

This release represents an update to the existing TAPS tools on a functional level, providing the same capabilities as the legacy TAPS desktop applications. However, these capabilities have been reimplemented using modern, cloud-ready software development techniques and frameworks. The internal structure of the application has been redesigned, streamlining and optimizing the system design and process flow, taking full advantage of cloud platform capabilities.

2.1 Deployment Environment

TAPS is hosted on the UC2S operational environment and is comprised of two OpenShift provisioned servers and an AWS RDS instance. Diagram below represents the TAPS infrastructure.

Architecture Diagram



2.2 Software Functionality and Capabilities

The TAPS tool utilizes information from the Aeronautical Application Suite (AAS) to prepopulate data into OpenCities to aid in Graphic Terminal rendering and production. The TAPS tool ingests a user-specified output file of DAFIF data, which includes Runways, ILS components, Nav aids, Waypoints, Heliports, SUAs, and Air Boundary areas. The features can be filtered via Country, Province, Airport, and a Search Radius. Data is ran and made available to users on a bi-weekly basis, but can be ran weekly or as needed dependent on users needs.

By default, users are not allowed access to the web application and/or API. When new users visit the site, they will be shown a configurable message telling how to acquire access. Administrators can grant users access to the application via the User Administration page on the website.

Administrators of the system can not only configure user access and permissions, but they can also upload new data sets and maintain other system configuration details.



2.3 Inventory of Software Components

This section documents the software configuration for TAPS.

A single web interface utilizing React serves as the user interface to the backend API developed as a cross-platform service using C# and .NET Core interacting with a modernized PostgreSQL database, which is the open source relational database management system. The TAPS Web Client Application Software consists of Node.js, the open source JavaScript runtime. The JavaScript dependencies and libraries are managed by Node.js and are embedded into the TAPS Web Client package files.

2.3.1 Software Deliverables

TAPS will deliver custom software packages and build artifacts containing the application components, including but not limited to .NET Dynamic Linked Library (DLL) files, JavaScript, HTML, and CSS scripts. These packages will include code library dependencies along with custom code to ensure that the application has all the software needed to execute.

2.3.2 Software List

The TAPS Version 1.0.0 Software Lists for custom code, COTS and Open Source code can be found in Table 2.3.2.

Table 2.3.2 - Software List

Software	Version	SWID	Vendor	Control #	Notes
Terminal Approach Production System	1.0.0	33969	Aeronautical Navigation Office (SFA)	202503240039H	
Pipeline					
Node.js 22	22.13.1	31768	OpenJS Foundation	202501240092H	
NPM Packages					
@emotion/react	11.14.0	30295	Mateusz Burzynski	202502040034H	
@emotion/styled	11.14.0	31193	Github/NPM	202501150103H	
@fontsource/roboto	5.1.1	32419	fontsource	202501150102H	
@mui/icons-material	6.4.4	24615	Material-UI	202502200038H	
@mui/material	6.4.4	24255	Olivier Tassinari	202502200039H	
@mui/x-data-grid	7.27.0	24262	Flavien DeLangle	202502200041H	
@mui/x-date-pickers	7.27.0	25233	Material-UI	202502200043H	
dayjs	1.11.13	21213	iamkun	20240827005	
notistack	3.0.2	22039	Hossein Dehnokhalaji	202502200056H	
prop-types	15.8.1	26615	Facebook	20230615069	
react	18.3.1	7572	Dan Abramov	20240521080	
react-dom	18.3.1	31097	Joseph Savona	20240521082	
react-draggable	4.4.6	19416	Samual Reed	20241007112	
react-hook-form	7.54.2	20912	Bill Lou	202501140089H	
react-input-mask	2.0.4	24992	Nikita Lobachev	20220311017	
react-number-format	5.4.3	21782	Sudhanshu Yadav	202501140086H	
react-router-dom	7.1.5	23883	Tim Dorr	202502200064H	
recoil	0.7.7	30077	Facebook	20231114040	



sass	1.85.0	20614	Natalie Weizenbaum	202502200065H	
whatwg-fetch	3.6.20	17984	Anne van Kersteren	20240402087	
.NET 8 Package Dependencies					
Autofac	8.1.1	25477	Autofac	20241106057	
Autofac.Extensions.DependencyInjection	10.0.0	25475	Autofac	20241003077	
AutoMapper	13.0.1	22280	Jimmy Bogard	20241106058	
DocumentFormat.OpenXml	3.1.1	27800	Nuget	20241106059	
Microsoft.AspNetCore.Authentication.JwtBearer	8.0.10	25083	Microsoft	20241106061	
Microsoft.AspNetCore.OpenApi	8.0.10	32236	Microsoft	20241106062	
Microsoft.AspNetCore.SpaServices.Extensions	8.0.10	27802	NuGet	20241106063	
Microsoft.EntityFrameworkCore	8.0.10	25412	Microsoft	20241106064	
Microsoft.EntityFrameworkCore.Design	8.0.10	25413	Microsoft	20241106065	
NetTopologySuite.IO.GeoJSON	4.0.0	27803	NuGet	20241007117	
Npgsql.NetTopologySuite	8.0.5	33398	NuGet	20241106069	
Npgsql.EntityFrameworkCore.PostgreSQL	8.0.10	27805	NuGet	20241106066	
Npgsql.EntityFrameworkCore.PostgreSQL.NetTopologySuite	8.0.10	27806	NuGet	20241106067	
Swashbuckle.AspNetCore	6.9.0	25175	Richard Morris	20241106068	
NPM Development Dependencies					
@babel/core	7.26.9	18857	Nicolo Ribaudo	202502200029H	
@babel/eslint-parser	7.26.8	21659	The Babel Team	202502200033H	
@babel/preset-env	7.26.9	18855	The Babel Team	202502200035H	
@babel/preset-react	7.26.3	19014	The Babel Team	202501300040H	
babel-plugin-transform-react-remove-prop-types	0.4.24	19058	Olivier Tassinari	20230522015	
@emotion/babel-plugin	11.13.5	30475	emotion-js	20241203006	
@testing-library/dom	10.4.0	21363	Kent C Dodds	202501270049H	
@testing-library/react	16.2.0	21140	Kent C Dodds	202502200045H	
@testing-library/user-event	14.6.1	21353	Kent C Dodds	202502200046H	
@types/react	18.3.4	18058	Nathan Shively-Sanders	20240912026	
@types/react-dom	18.3.0	18053	Nathan Shively-Sanders	20240606089	
@vitejs/plugin-react	4.3.4	30081	Vite	202501150091H	
babel-loader	9.2.1	15661	Daniel Tschinder	20240917005	
eslint	9.20.1	15393	Nicholas Zakas	202502200015H	
eslint-config-prettier	10.0.1	17083	Simon Lydell	202501210034H	
eslint-plugin-import	2.31.0	15560	Jordan Harband	20241008017	
eslint-plugin-jsdoc	50.6.3	18977	Brett Zamir	20250211056H	
eslint-plugin-jsx-a11y	6.10.2	15587	Ethan Cohen	202501300049H	
eslint-plugin-react	7.37.4	15721	Jordan Harband	202501210035H	
eslint-plugin-react-hooks	5.1.0	19480	Facebook	202501300048H	
eslint-plugin-react-refresh	0.4.19	33109	Amaud Barre	202502200016H	
eslint-plugin-security	3.0.1	19592	Node Security	20240807029	
eslint-plugin-unused-imports	4.1.4	30477	TypeScript ESLint	202502200054H	
globals	15.15.0	32483	Sindre Sorhus	202502200017H	
husky	9.1.7	21706	typicode	202502200055H	
prettier	3.5.1	26610	Fisker Cheung	202502200057H	
vite	6.1.0	26181	Evan You	202502200022H	



vite-plugin-checker	0.8.0	30474	Vite	202502200066H	
vitest	3.0.6	30678	Vladimir Rutsky	202502200067H	
web-vitals	4.2.4	24705	Philip Walton	202501230031H	
.NET 8 Development Dependencies					
coverlet.collector	6.0.2	29679	Toni Solarin-Sodara	20241119050	
JUnitXml.TestLogger	4.1.0	33470	NuGet	20241114130	
Microsoft.EntityFrameworkCore.InMemory	8.0.10	29671	Microsoft	20241119054	
Microsoft.NET.Test.Sdk	17.11.1	20204	Microsoft	20241114132	
Moq	4.20.72	32811	NuGet	20241114133	
NUnit	4.2.2	20340	Charlie Poole	20241114134	
NUnit.Analyzers	4.3.0	33469	mikkelbu	20241114137	
NUnit3TestAdapter	4.6.0	20341	Charlie Poole	20241114138	
Verify.NUnit	27.0.1	33468	simoncropp	20241119064	

2.4 Changes Installed

As was noted in Section 1.3, this is considered to be in initial release of the 1.0.0 TAPS product so changes from the previous version are not recorded in this VDD.

Table 2.4 includes defects recorded in testing as well as any other internally discovered defects that were resolved.

- Issue Types:
- Test Problem Report (TPR): DRs written by the NTO during testing
- System Problem Report (SPR): DRs written by problems found by the PMO, customer, or external systems.

Table 2.4 - Changes Installed

PBI/INC #	Key	Issue Type	Summary

2.5 Installation Instructions

The application components will be installed using the NGA CORE services GitLab CI/CD pipeline. No special install instructions will be needed once the pipeline is set up and configured.

(Instructions on how to manually install and configure the application, if needed for this document, are To Be Determined).

2.6 Possible Problems and Known Errors

Table 2.6 Version 6.0.0 Possible Problems and Known Errors

Issue ID	Summary	Description